

表 1: Performance on Kinetics-400 in terms of Top-1/5 Accuracy (%). Red rows indicate the gain compared to the baseline.

Method	Type	TPN $\rightarrow$ SlowFast		VideoST $\rightarrow$ SlowFast	
		Top1 $\uparrow$	Top5 $\uparrow$	Top1 $\uparrow$	Top5 $\uparrow$
-	Teac.	63.23	84.46	74.07	91.05
	Stud.	51.08	76.05	51.08	76.05
KD [2]	arXiv'15	52.41	77.12	53.62	78.45
FitNet[6]	ICLR'15	53.15	77.85	54.28	79.12
DKD [11]	CVPR'22	56.82	80.34	62.45	84.67
CTKD[5]	AAAI'23	57.54	80.92	64.12	85.33
GKD[8]	AAAI'24	59.12	81.67	67.85	87.42
CrossKD [9]	CVPR'24	60.05	82.41	69.34	88.21
DualKD [10]	TIP'24	59.88	82.15	68.92	87.98
DCSF-KD [1]	AAAI'25	61.23	83.14	71.12	89.45
SAKD[4]	MM'25	<u>62.15</u>	<u>83.89</u>	<u>72.58</u>	<u>90.32</u>
Ours		62.87	84.12	73.65	90.84

表 2: Performance on Sth-Sthv2 in terms of Top-1/5 Accuracy (%).

Method	Type	TPN $\rightarrow$ SlowFast		VideoST $\rightarrow$ SlowFast	
		Top1 $\uparrow$	Top5 $\uparrow$	Top1 $\uparrow$	Top5 $\uparrow$
-	Teac.	55.10	83.92	58.17	86.28
	Stud.	43.96	71.70	43.96	71.70
KD [2]	arXiv'15	45.12	73.05	46.28	74.15
FitNet[6]	ICLR'15	45.85	73.68	47.02	74.92
DKD [11]	CVPR'22	49.32	78.14	51.45	80.36
CTKD[5]	AAAI'23	50.15	78.96	52.88	81.45
GKD[8]	AAAI'24	51.74	80.22	54.32	82.91
CrossKD [9]	CVPR'24	52.68	81.15	55.60	83.84
DualKD [10]	TIP'24	52.45	80.92	55.18	83.52
DCSF-KD [1]	AAAI'25	53.52	82.04	56.45	84.72
SAKD[4]	MM'25	<u>54.28</u>	<u>82.87</u>	<u>57.21</u>	<u>85.45</u>
Ours		54.82	83.56	57.94	86.03

1 Experiments

1.1 Quantitative Results

实验结果表明，本方法在 Kinetics-400 和 Sth-Sthv2 两个核心数据集上均展现出卓越的知识迁移能力。在侧重空间语义的 Kinetics-400 任务中，本方法将学生模型的 Top-1 准确率提升至 62.87% 和 73.65%，与教师模型的性能差距减少至 1% 以内，显著优于 SAKD 等目前最新方法；而在更依赖时序建模的 Sth-Sthv2 任务上，本方法的表现提升突出，将学生模型从 43.96% 的准确率提升至 54.82%，甚至在 VideoST $\rightarrow$ SlowFast 实验中准确率达到 57.94%，均与原先的教师模型差距不到 1%。

在第一阶段中，采用了加权特征损失 L_{fea}^{kd1} ，并在表3中对比了不同的对齐策略。第 1 行作为基准

表 3: Model performance of distillation method with Stage 1 align math on UCF101, Kinetics-400, and Sth-Sthv2. Red rows indicate the gain compared to the baseline Align(1*1cov).

Align	UCF101[7]				Kinetics-400[3]				Sth-Sthv2			
	Top1↑		Top5↑		Top1↑		Top5↑		Top1↑		Top5↑	
	SlowFast	Video ST	SlowFast	Video ST	SlowFast	Video ST	SlowFast	Video ST	SlowFast	Video ST	SlowFast	Video ST
Align(1*1cov)	85.23	86.27	95.89	96.63	55.64	72.81	81.23	91.28	45.62	73.94	54.39	84.22
Attention	85.38	86.99	96.10	96.78	56.02	73.19	<u>81.89</u>	91.83	45.98	74.21	54.88	84.65
Gain	+0.15	+0.72	+0.21	+0.15	+0.38	+0.38	+0.66	+0.55	+0.36	+0.27	+0.49	+0.43
Frequency Align	87.29	87.93	97.31	97.23	58.32	74.38	82.93	91.73	46.45	75.12	56.12	85.80
Gain	+2.06	+1.66	+1.42	+0.60	+2.68	+1.57	+1.70	+0.45	+0.83	+1.18	+1.73	+1.58
Numerator	87.38	88.12	<u>97.52</u>	<u>97.82</u>	59.38	75.02	81.76	<u>92.38</u>	<u>46.91</u>	<u>75.66</u>	56.84	86.15
Gain	+2.15	+1.85	+1.63	+1.19	+3.74	+2.21	+0.53	+1.10	+1.29	+1.72	+2.45	+1.93
Ours	88.42	<u>88.04</u>	97.80	98.07	60.23	75.83	82.38	93.22	47.28	76.03	57.31	86.87
Gain	+3.19	+1.77	+1.91	+1.44	+4.59	+3.02	+1.15	+1.94	+1.66	+2.09	+2.92	+2.65

(Baseline)，仅使用 1×1 卷积进行通道对齐。在此基础上，第 2 行加入了两个注意力层以增强教师与学生模型间的空间特征对齐。第 3 行转而在频域进行特征对齐，第 4 行则将质心频率差 (Centroid Frequency Difference) 直接作为对齐权重。在 UCF101 数据集上，相比于基准，我们的方法 (Ours) 通过采用中心频率差的倒数作为特征 MSE 损失的权重，实现了约 1.77% 至 3.19% 的显著性能提升。在 Kinetics-400 数据集上，频域对齐 (第 3 行) 表现出明显优于空间对齐方法 (第 1 和 2 行) 的增益，而分子加权 (第 4 行) 在频域对齐的基础上进一步优化了结果。最后，在 Sth-Sthv2 数据集上，本文的策略再次超越了所有对比方法，最高带来了 2.09% 和 2.65% 的 Top-1/5 精度增长。Ours 方案采用了倒数加权策略，这充分考虑了训练初期中心频率不稳定的事实，在频率差异较大时适度抑制特征对齐的权重，从而避免了无效的监督信号对训练的干扰，最终在三个数据集上均取得了最佳性能。

参考文献

- [1] Tao Dai, Yang Lin, Hang Guo, Jinbao Wang, and Zexuan Zhu. DCSF-KD: dynamic channel-wise spatial feature knowledge distillation for object detection. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, pages 2627–2635, 2025.
- [2] Oriol Vinyals Geoffrey Hinton and Jeffrey Dean. Distilling the knowledge in a neural network. *arXiv preprint*, arXiv:1503.02531, 2015.
- [3] Will Kay, Joao Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, Mustafa Suleyman, and Andrew Zisserman. The kinetics human action video dataset. *arXiv preprint arXiv:1705.06950*, 2017.
- [4] Ping Li, Chenhao Ping, Wenxiao Wang, and Mingli Song. Sample-level adaptive knowledge distillation for action recognition. *Association for Computing Machinery (ACM MM)*, pages 544–552, 2025.
- [5] Zheng Li, Xiang Li, Lingfeng Yang, Borui Zhao, Renjie Song, Lei Luo, Jun Li, and Jian Yang. Curriculum

- temperature for knowledge distillation. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, pages 1504–1512, 2023.
- [6] Adriana Romero, Nicolas Ballas, Samira Ebrahimi Kahou, Antoine Chassang, Carlo Gatta, and Yoshua Bengio. Fitnets: Hints for thin deep nets. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2015.
- [7] Khurram Soomro, Amir Roshan Zamir, and Mubarak Shah. Ucf101: A dataset of 101 human actions classes from videos in the wild. *arXiv preprint arXiv:1212.0402*, 2012.
- [8] Guiqin Wang, Peng Zhao, Yanjiang Shi, Cong Zhao, and Shusen Yang. Generative model-based feature knowledge distillation for action recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, pages 15474–15482, 2024.
- [9] Jiabao Wang, Yuming Chen, Zhaohui Zheng, Xiang Li, Ming-Ming Cheng, and Qibin Hou. Crosskd: Cross-head knowledge distillation for object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16520–16530, 2024.
- [10] Rui Wang, Yixue Hao, Long Hu, Xianzhi Li, Min Chen, Yiming Miao, and Iztok Humar. Efficient crowd counting via dual knowledge distillation. *IEEE Transactions on Image Processing (TIP)*, 33:569–583, 2024.
- [11] Borui Zhao, Quan Cui, Renjie Song, Qiu Yiyu, and Liang Jiajun. Decoupled knowledge distillation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11953–11962, 2022.